

Exercise Solutions of Chp 3 N. Goldenfeld

Phase T. and RN

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1 Chapter 3

1.1 Exercise 3.1

Transfer matrix in $d = 1$ Ising model with *periodic* boundary conditions:

(a) We construct the matrix S which diagonalises the transfer matrix T such that

$$T' = S^{-1}TS \quad (1.1)$$

is diagonal: Recall $h := \beta H$, $K := \beta J$ and

$$T = \begin{pmatrix} e^{h+K} & e^{-K} \\ e^{-K} & e^{-h+K} \end{pmatrix}. \quad (1.2)$$

Define now the e-vals with the diagonal matrix

$$T' = \begin{pmatrix} \lambda_+ & \\ & \lambda_- \end{pmatrix}. \quad (1.3)$$

and it can easily be seen that

$$\lambda^2 - 2\lambda e^K \cosh(h) + 2 \sinh(2K) = 0, \quad (1.4)$$

where $\lambda \sim \lambda_{(\pm)}$. It follows then that

$$\lambda = e^K [\cosh(h) \pm \sqrt{\sinh^2(h) + e^{-4K}}]. \quad (1.5)$$

To diagonalize the matrix we must find e-vectors which involves solving

$$T\vec{v} = \lambda_{(\pm)}\vec{v}. \quad (1.6)$$

Let $\vec{v}_{\pm} = (a_{\pm}, b_{\pm})$. Then, if we define

$$\cot(2\phi) = e^{2K} \sinh(h), \quad (1.7)$$

and plug away at a lot of algebra, we get to

$$\begin{aligned} b_+ &= a_+ \tan(\phi) \\ a_- &= -b_- \cot(\phi) \end{aligned} \quad (1.8)$$

if we require our e-vectors to be normalized this gives us

$$\begin{aligned} \vec{v}_+ &= (\cos(\phi), \sin(\phi)), \\ \vec{v}_- &= (\sin(\phi), -\cos(\phi)), \end{aligned} \quad (1.9)$$

so

$$S = \begin{pmatrix} \cos(\phi) & \sin(\phi) \\ \sin(\phi) & -\cos(\phi) \end{pmatrix} \quad (1.10)$$

where

$$\cot(2\phi) = e^{2K} \sinh(h). \quad (1.11)$$

(b)

Note that

$$S^{-1} = S \quad (1.12)$$

So that

$$S\sigma_z S^{-1} = \begin{pmatrix} \cos^2(\phi) - \sin^2(\phi) & 2\cos(\phi)\sin(\phi) \\ 2\cos(\phi)\sin(\phi) & -(\cos^2(\phi) - \sin^2(\phi)) \end{pmatrix} = \begin{pmatrix} \cos(2\phi) & \sin(2\phi) \\ \sin(2\phi) & -\cos(2\phi) \end{pmatrix} \quad (1.13)$$

Thus

$$S\sigma_z S^{-1}(T')^N = \begin{pmatrix} \cos(2\phi) & \sin(2\phi) \\ \sin(2\phi) & -\cos(2\phi) \end{pmatrix} \begin{pmatrix} \lambda_1^N & \\ & \lambda_2^N \end{pmatrix} = \begin{pmatrix} \lambda_1^N \cos(2\phi) & \lambda_2^N \sin(2\phi) \\ \lambda_1^N \sin(2\phi) & -\lambda_2^N \cos(2\phi) \end{pmatrix} \quad (1.14)$$

Recall also that

$$\langle S_i \rangle = \frac{\text{Tr}(S^{-1}\sigma_z S(T')^N)}{Z_n}, \quad (1.15)$$

where $Z_N = \lambda_1^N + \lambda_2^N$. THus we have

$$\langle S_i \rangle = \frac{\lambda_1^N \cos(2\phi) - \lambda_2 \cos(2\phi)}{\lambda_1 + \lambda_2} = \frac{\cos(2\phi) - (\lambda_2/\lambda_1)^N \cos(2\phi)}{1 + (\lambda_2/\lambda_1)^N}. \quad (1.16)$$

Taking $N \rightarrow \infty$ this becomes

$$\langle S_i \rangle = \cos(2\phi). \quad (1.17)$$

As shown in the book

$$\langle S_i S_{i+j} \rangle 1/Z \text{Tr}(S^{-1}\sigma_z S(T')^j S^{-1}\sigma_z S(T')^{N-j}). \quad (1.18)$$

Define

$$S^{-1}\sigma_z S = \begin{pmatrix} e & g \\ f & k \end{pmatrix} \quad (1.19)$$

Then as $N \rightarrow \infty$

$$\langle S_i S_{i+j} \rangle = e^2 + gf \left(\frac{\lambda_2}{\lambda_1}\right)^j = \cos^2(2\phi) + \sin^2(2\phi) \left(\frac{\lambda_2}{\lambda_1}\right)^j \quad (1.20)$$

showing that the system is translationally invariant and

$$G(i, i+j) = \langle S_i S_{i+j} \rangle - \langle S_i \rangle \langle S_j \rangle = \sin^2(2\phi) \left(\frac{\lambda_2}{\lambda_1}\right)^j \quad (1.21)$$

(c) Recall

$$M = \sinh(h)/\sqrt{\sinh^2(h) + e^{-4k}} \quad (1.22)$$

or alternatively

$$M(H) = \sinh(\beta H)/\sqrt{\sinh^2(\beta H) + e^{-4k}} \quad (1.23)$$